Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay_compression.
- Best TSPLIB holdout gap: tsplib_failure_replay_compression.
- Best synthetic holdout gap: tsplib_random_replay.

Run Metadata

- run_name: run_20260513_130152_c
- started_at_local: 2026-05-13 13:01:52
- finished_at_local: 2026-05-13 13:12:39
- duration_hhmm: 00:11
- duration_seconds: 647.091
- seed_offset: 2000
- replicate_label: c
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.819021	0.56	-0.030408
tsplib_random_replay	random	score_only	False	0.15793	0.0	0.100501	0.815172	0.76	0.126262
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.892558	0.58	-0.027903
tsplib_failure_replay_compression	failure	novelty_gate	True	0.092257	0.0	0.058709	0.835391	0.76	0.075976

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.819021, and final complexity 0.56.
- Adaptation efficiency -0.030408 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.15793, synthetic 0.0, combined 0.100501.
- Accepted-epoch count 2, mean accepted code novelty 0.815172, and final complexity 0.76.
- Adaptation efficiency 0.126262 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.15793 across 7 instances; family means: ch=0.179697, kroD=0.185921, pcb=0.215014, pr=0.106399, rd=0.121745, st=0.117037.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3540, "family": "a", "name": "a280", "optimality_gap": 0.372625}, {"best_known_cost": 629, "cost": 828, "family": "eil", "name": "eil101", "optimality_gap": 0.316375}, {"best_known_cost": 14379, "cost": 17442, "family": "lin", "name": "lin105", "optimality_gap": 0.213019}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.892558, and final complexity 0.56.
- Adaptation efficiency -0.027903 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.092257, synthetic 0.0, combined 0.058709.
- Accepted-epoch count 2, mean accepted code novelty 0.835391, and final complexity 0.76.

- Adaptation efficiency 0.075976 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.092257 across 7 instances; family means: ch=0.077617, kroD=0.108904, pcb=0.203828, pr=0.119777, rd=0.03287, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3449, "family": "a", "name": "a280", "optimality_gap": 0.33734}, {"best_known_cost": 629, "cost": 734, "family": "eil", "name": "eil101", "optimality_gap": 0.166932}, {"best_known_cost": 14379, "cost": 16255, "family": "lin", "name": "lin105", "optimality_gap": 0.130468}]

Judge Appendix

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Replay-Aware TSP Benchmark Suite Analysis

Conditions Overview

- **No replay** (tsplib_no_replay): Replay mode = none, selection = score_only
- **Random replay** (tsplib_random_replay): Replay mode = random, selection = score_only
- **Failure replay** (tsplib_failure_replay): Replay mode = failure, selection = score_only
- **Failure replay with compression** (tsplib_failure_replay_compression): Replay mode = failure, selection = novelty_gate, compression applied

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Key Metrics for Transfer (Held-out TSPLIB & Synthetic Holdout)

Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Replay Mode	Selection Mode
tsplib_no_replay	0.213387	0.064916	0.159398	0.819021	none	score_only
tsplib_random_replay	0.15793	0.0	0.100501	0.815172	random	score_only
tsplib_failure_replay	0.213387	0.064916	0.159398	0.892558	failure	score_only
tsplib_failure_replay_compression	**0.092257**	**0.0**	**0.058709**	0.835391	failure	novelty_gate

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Interpretation

1. **Optimality gaps indicate best transfer with tsplib_failure_replay_compression:**

- Lowest mean TSPLIB gap (0.092257) compared to all other conditions.

- Zero synthetic holdout gap, matching or outperforming others.
- Lowest mean transfer gap (0.058709).

2. **Random replay improves transfer over no replay despite slight code novelty reduction:**
- tsplib_random_replay has lower TSPLIB and transfer gaps than no replay.
- Code novelty slightly lower than no replay (0.815 vs 0.819).
- This shows improved transfer not driven by higher code novelty or algorithmic novelty.

3. **Failure replay without compression does not improve transfer over no replay:**
- Has identical TSPLIB and transfer gaps as no replay condition.
- However, this condition shows increased code novelty (~0.89 vs ~0.82).
- Thus, increased code novelty here did not yield better transfer.

4. **Compression-aware failure replay enhances transfer and lowers optimality gaps, with modest code novelty:**
- Despite slightly lower code novelty (0.835 vs 0.89 in failure replay), this condition achieves the best transfer metrics.
- Use of novelty gating for selection and compression pressure likely contribute to improved performance.

5. **Behavior profiles and complexity:**
- Conditions with better transfer (random_replay and failure_replay_compression) have higher final complexity (0.76) vs others (0.56), possibly indicating more sophisticated strategies.
- failure_replay_compression uses novelty gating, which differs from other score-only selection modes, possibly contributing to improvements.

Summary

- **Best transfer performance is achieved by tsplib_failure_replay_compression**, with the lowest optimality gaps on held-out TSPLIB and synthetic benchmarks.
- **Random replay condition improves transfer over no replay despite slightly lower code novelty**, showing that increased transfer is not always linked with code novelty spikes.
- **Failure replay without compression did not improve transfer despite higher code novelty**, indicating lexical novelty alone does not imply algorithmic improvement.
- **Compression-aware replay with novelty-based selection enhances transfer**, outperforming other replay modes.
- Careful distinction between replay types and use of compression-aware strategies is critical for improved generalization and optimality gap reduction.

No evidence from optimality-gap or transfer data suggests that lexical code novelty equates to algorithmic invention here. Improvements in transfer correlate more with replay type and compression-aware selection than with higher code novelty alone.